

DecodeLabs – Data Analytics Project 2

Exploratory Data Analysis (EDA) Report

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Project Title

Exploratory Data Analysis of an E-Commerce Sales Dataset

1. Introduction

The objective of this project was to perform Exploratory Data Analysis (EDA) on an e-commerce sales dataset. The analysis aimed to identify sales patterns, customer behavior, revenue trends, order performance, and relationships among important variables. Various statistical and visualization techniques were used to gain meaningful business insights.

2. Dataset Overview

The dataset contains transactional information from an e-commerce platform.

Dataset Summary

- Total Records: 1200
- Total Columns: 14

Variables Included

- OrderID
- Date
- CustomerID
- Product
- Quantity
- UnitPrice
- ShippingAddress
- PaymentMethod

- OrderStatus
- TrackingNumber
- ItemsInCart
- CouponCode
- ReferralSource
- TotalPrice

3. Data Cleaning and Preparation

Before analysis, the dataset was examined for quality issues.

Steps Performed

- Checked for missing values
- Checked for duplicate records
- Verified numerical and categorical variables
- Created month-wise analysis fields
- Prepared data for Pivot Tables and correlation analysis

Findings

- No major missing values were identified.
- No significant duplicate records were found.
- The dataset was suitable for analysis.

4. Descriptive Statistics

The following numerical variables were analyzed:

- Quantity
- UnitPrice
- ItemsInCart
- TotalPrice

Measures such as Mean, Median, Maximum, Minimum, and Count were calculated to understand the overall distribution of sales data.

5. Product Analysis

Product-wise sales and revenue were analyzed using Pivot Tables and charts.

Findings

- Different products contributed differently to sales volume and revenue.
- A small number of products generated a significant portion of total sales.
- Product performance can be used for inventory and marketing decisions.

Visualization

- Product-wise Sales Chart
- Product-wise Revenue Chart

6. Payment Method Analysis

Payment method usage was analyzed to understand customer preferences.

Order Count by Payment Method

Payment Method	Orders
Online	258
Cash	246
Credit Card	234
Debit Card	232
Gift Card	230

Revenue by Payment Method

Payment Method	Revenue
Credit Card	263,847.63
Online	262,442.94
Cash	259,762.29
Gift Card	246,323.92
Debit Card	232,361.18

Key Findings

- Online payment was the most frequently used payment method with 258 orders.

- Credit Card generated the highest revenue (263,847.63).
- Debit Card generated the lowest revenue among payment methods.

7. Referral Source Analysis

Referral sources were analyzed to determine which marketing channels attracted the most customers.

Order Count by Referral Source

Referral Source	Orders
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

Revenue by Referral Source

Referral Source	Revenue
Instagram	257,258.45
Email	262,008.55
Google	250,441.48
Facebook	250,410.90
Referral	226,815.58

Key Findings

- Instagram generated the highest number of orders.
- Email marketing generated the highest revenue.
- Referral sources contributed the least revenue.

8. Monthly Sales Trend Analysis

Monthly revenue analysis was conducted to identify seasonal trends.

Revenue by Month

Month	Revenue
June	170,616.13
May	135,142.59
January	124,313.23
March	123,840.93
February	112,344.78
April	109,186.05
October	89,834.82
August	86,343.21
July	85,784.64
December	82,540.50
November	75,493.43
September	69,321.65

Key Findings

- June recorded the highest sales revenue.
- September recorded the lowest sales revenue.
- The first half of the year performed better than the later months.

9. Outlier Analysis

Box-and-Whisker plots were used to detect outliers in TotalPrice and Quantity.

Findings

- Several transactions showed exceptionally high order values.
- The maximum observed order value exceeded 3,400.
- Some transactions were significantly larger than average orders.
- These outliers may represent bulk purchases or premium product transactions.

10. Correlation Analysis

Correlation analysis was performed between Quantity, UnitPrice, ItemsInCart, and TotalPrice.

Correlation Matrix Results

Variables	Correlation
Quantity ↔ TotalPrice	0.615
UnitPrice ↔ TotalPrice	0.717
ItemsInCart ↔ TotalPrice	0.393
Quantity ↔ ItemsInCart	0.650
Quantity ↔ UnitPrice	0.015
UnitPrice ↔ ItemsInCart	0.001

Interpretation

- UnitPrice and TotalPrice showed a strong positive correlation (0.717).
- Quantity and TotalPrice also demonstrated a strong positive relationship (0.615).
- Quantity and ItemsInCart showed a moderate positive correlation (0.650).
- UnitPrice and ItemsInCart showed almost no relationship.
- UnitPrice is the strongest contributor to TotalPrice.

11. Key Insights

1. The dataset contained 1,200 customer transactions.
2. Online payment was the most commonly used payment method.
3. Credit Card payments generated the highest revenue.
4. Instagram attracted the highest number of customers.
5. Email marketing generated the highest revenue.
6. June was the best-performing month in terms of sales.
7. September generated the lowest monthly revenue.
8. Several high-value transactions were identified as outliers.
9. UnitPrice had the strongest impact on TotalPrice.
10. Quantity sold also played a significant role in revenue generation.

12. Conclusion

The Exploratory Data Analysis successfully revealed important patterns in customer purchasing behavior, sales performance, payment preferences, and marketing effectiveness. The analysis showed that payment methods, referral channels, product pricing, and sales volume significantly influenced revenue generation. Correlation analysis confirmed that UnitPrice and Quantity are the strongest drivers of TotalPrice. These findings can help businesses improve marketing strategies, optimize inventory management, and increase overall profitability.

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